DOCUMENT 2: DATASET DESIGN & METADATA DOCUMENT

NexaChain Intelligence Platform | GlobeNex Manufacturing & Logistics Corp.

5 Core Enterprise Datasets | 310,000+ Total Records | 150+ Columns

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DATASET PORTFOLIO OVERVIEW

Dataset	File Name	Total Rows	Total Columns	Primary Key	Domain Coverage
Dataset 1: Orders	orders.csv	120,000+	28 columns	order_id	Orders, Products, Customers, Revenue, Fulfillment, Profitability
Dataset 2: Inventory	inventory.csv	55,000+	26 columns	inventory_id	Stock Levels, Warehouses, Movements, Reorder, Holding Costs, Stock Age
Dataset 3: Vendors	vendors.csv	22,000+	30 columns	vendor_record_id	Vendor Info, Performance, Risk, Procurement, Quality, Lead Times
Dataset 4: Logistics	logistics.csv	85,000+	32 columns	shipment_id	Shipments, Transportation, Delivery, Fuel, Routes, Delays, Weather
Dataset 5: Financials	financials.csv	65,000+	28 columns	finance_record_id	Revenue, Procurement Cost, OpEx, Working Capital, Cash Flow, Risk

Dataset Relationships Overview

All 5 datasets are interconnected through a set of shared foreign keys that simulate real enterprise data relationships. The Orders dataset serves as the central fact table, linking to Inventory (via product_id), Vendors (via vendor_id), Logistics (via order_id / shipment_id), and Financials (via order_id and vendor_id). This relational structure enables cross-dataset joins for multi-domain analytics.

Source Dataset	Target Dataset	Join Key	Relationship Type
Orders	Logistics	order_id	One-to-Many (1 order may have multiple shipments)
Orders	Inventory	product_id	Many-to-Many (orders contain products; products tracked in inventory)

Orders	Vendors	vendor_id	Many-to-One (multiple orders from same vendor)
Orders	Financials	order_id	One-to-One (each order has a financial record)
Vendors	Financials	vendor_id	One-to-Many (vendor appears in multiple financial transactions)
Logistics	Financials	shipment_id	One-to-One (each shipment has cost records in Financials)
Inventory	Vendors	vendor_id	Many-to-One (inventory items sourced from vendors)

DATASET 1: ORDERS.CSV — COMPLETE METADATA

Dataset Name	Orders Dataset
File Name	orders.csv
Total Rows	120,000+ records (synthetic, enterprise-grade)
Total Columns	28 columns
Date Range	January 1, 2021 to December 31, 2024 (4 years)
Primary Key	order_id — unique alphanumeric order identifier (format: ORD-XXXXXXXX)
Foreign Keys	vendor_id (→ Vendors), product_id (→ Inventory), customer_id
Business Purpose	Central transaction table capturing all customer orders with product, vendor, delivery, fulfillment, and profitability attributes. Serves as the primary fact table for cross-dataset analysis.
Grain	One row per order line item (a single order_id may have multiple rows if multi-product)
Update Frequency	Daily batch load from SAP S/4HANA order management module
Data Quality Issues Introduced	~3% duplicate order_id entries, ~5% NULL values in delivery_date and profit_margin, ~2% negative order_value entries (data entry errors), ~4% future-dated order dates, ~3% inconsistent status categories (e.g., 'Shipped' vs 'SHIPPED' vs 'shipped'), ~1% product_id referencing non-existent inventory records

Column Data Dictionary — Orders.csv

All 28 columns are documented below with complete metadata for intern reference and data governance compliance.

Column Name	Description	Datatype	Business Meaning	Allowed Values	Validation Rules	Null Handling
order_id	Unique order identifier	VARCHAR(15)	Primary key for order tracking	ORD-XXXXXXXX format	Must be unique, non-null, format validated	NOT NULL — reject record
order_date	Date order was placed	DATE	Order creation timestamp	2021-01-01 to 2024-12-31	Must be valid date, not future	NOT NULL — flag for review
customer_id	Customer unique ID	VARCHAR(12)	Links order to customer master	CUST-XXXXXX format	Must exist in customer reference	NOT NULL — reject record

customer_name	Customer company name	VARCHAR(100)	Client identification for reporting	Any string	Non-empty string	NULL allowed — flag only
customer_country	Country of customer	VARCHAR(50)	Geographic segmentation	ISO 3166 country names	Must be valid country	NOT NULL — default 'Unknown'
customer_segment	Customer tier/segment	VARCHAR(30)	Segmentation for analytics	Enterprise, Mid-Market, SMB, Government	Must match allowed values	NOT NULL — flag
product_id	Product unique identifier	VARCHAR(12)	Links to Inventory dataset	PRD-XXXXXX format	Must exist in inventory.csv	NOT NULL — reject record
product_name	Product description	VARCHAR(150)	Human-readable product label	Any string	Non-empty	NOT NULL
product_category	Product classification	VARCHAR(50)	Category analytics and filtering	Industrial, Precision, Logistics, Electronics, Raw Materials	Must match master list	NOT NULL — flag
vendor_id	Supplying vendor ID	VARCHAR(12)	Links to Vendors dataset	VND-XXXXXX format	Must exist in vendors.csv	NOT NULL — reject record
order_quantity	Units ordered	INTEGER	Volume metric for demand	1 to 50,000	Positive integer only	NOT NULL — reject record
unit_price_usd	Price per unit (USD)	DECIMAL(10,2)	Revenue calculation base	0.01 to 99,999.99	Positive, 2 decimal places	NOT NULL — reject record
order_value_usd	Total order value (USD)	DECIMAL(12,2)	Revenue recognition amount	qty * unit_price	Must equal qty * unit_price (±0.01)	NOT NULL — recalculate if NULL
discount_pct	Discount applied (%)	DECIMAL(5,2)	Margin impact from discounts	0.00 to 40.00	0-40% range only	NULL = 0% (no discount)
cost_of_goods_usd	COGS for this order line	DECIMAL(12,2)	Gross margin calculation	Positive, < order_value	Must be less than order_value	NOT NULL — flag

gross_margin_usd	Gross profit (USD)	DECIMAL(12,2)	Profitability metric	order_value - COGS	Can be negative (loss orders)	Derived field — recalculate
profit_margin_pct	Gross margin (%)	DECIMAL(5,2)	Margin percentage for analytics	-100 to 100	Flag if < -20% or > 60%	5% NULL rate — expected
order_status	Current order lifecycle stage	VARCHAR(30)	Fulfillment tracking	Pending, Confirmed, Processing, Shipped, Delivered, Cancelled, Returned	Normalize case — known issue	NOT NULL — reject record
promised_delivery_date	Committed delivery date to customer	DATE	SLA baseline for OTD calculation	Order date + 3 to 120 days	Must be >= order_date	NOT NULL — flag
actual_delivery_date	Date order was actually delivered	DATE	Delivery performance measurement	Any valid date	NULL for non-Delivered orders	NULL if status != Delivered — OK
delivery_delay_days	Days late (positive = late)	INTEGER	Delay analytics; derived metric	-30 to 90	Negative = early delivery	Recalculate from dates if NULL
fulfillment_channel	Order fulfillment method	VARCHAR(30)	Channel analytics	Direct, Distributor, Drop-Ship, 3PL	Must match allowed values	NOT NULL — default 'Direct'
warehouse_id	Dispatching warehouse ID	VARCHAR(10)	Links to warehouse master	WH-XX format, 14 warehouses	Must match warehouse list	NULL for Drop-Ship orders — OK
shipment_id	Links to Logistics dataset	VARCHAR(15)	Cross-dataset join key	SHP-XXXXXXXX format	Must exist in logistics.csv if not NULL	NULL for Pending/Cancelled — OK
return_reason	Reason if order returned	VARCHAR(100)	Returns analytics	Quality, Damage, Wrong Item, Delay, Other	Only populated if Returned	NULL if not Returned — OK
region	Sales region classification	VARCHAR(30)	Geographic business analysis	APAC, EMEA, North America, South America, Africa	Must match master list	NOT NULL — derive from country

created_by	User who created the order	VARCHAR(50)	Audit trail	Employee ID or System ID	Non-empty string	NOT NULL
last_modified_date	Last update timestamp	DATETIME	Data lineage tracking	Valid datetime	Must be >= order_date	NOT NULL — use created date

Data Quality Issues — Orders.csv

Issue Type	Description	Estimated Prevalence	Remediation Approach
Duplicates	Duplicate order_id values with slightly different timestamps — simulates ERP double-posting from AS/400 to SAP reconciliation	~3,600 records (3%)	Deduplicate on order_id keeping latest last_modified_date; flag all duplicates in a cleaning log
NULL Values	delivery_date and profit_margin_pct have intentional NULLs — delivery_date is NULL for non-Delivered orders (expected) and randomly for Delivered (unexpected)	~6,000 records (5%)	For expected NULLs: document in data dict. For unexpected: impute using model or flag for source-system correction
Negative Values	order_value_usd has negative entries from credit notes and data entry errors mixed into main orders table without proper type flag	~2,400 records (2%)	Separate credit notes via is_credit_note flag; negative values without flag = data entry error, set to NULL and flag
Future Dates	order_date values in 2025-2026 range — from batch processing timestamps incorrectly written to order_date field	~4,800 records (4%)	Cap at max valid date (2024-12-31); log records in data quality report for source system correction
Case Inconsistency	order_status field has mixed cases: 'Shipped', 'SHIPPED', 'shipped', 'Ship', 'ship.' — from multiple source system integrations	~3,600 records (3%)	Standardize using pd.str.strip().str.title() then map to canonical values; document mapping table
Referential Integrity	~1% of product_id values do not match any record in inventory.csv — orphaned records from discontinued products	~1,200 records (1%)	Flag as orphaned; join to last-known inventory record using fuzzy product name match; exclude from ML training

DATASET 2: INVENTORY.CSV — COMPLETE METADATA

Dataset Name	Inventory Dataset
File Name	inventory.csv
Total Rows	55,000+ records
Total Columns	26 columns
Date Range	January 2021 to December 2024 (daily snapshot records)
Primary Key	inventory_id — format: INV-XXXXXXXXXX
Foreign Keys	product_id (→ Orders), vendor_id (→ Vendors), warehouse_id
Business Purpose	Tracks daily inventory snapshots across 14 warehouses for all active SKUs. Includes stock levels, reorder metrics, holding costs, safety stock calculations, and stock age classification. Critical for inventory optimization and stockout prediction models.
Grain	One row per product-warehouse-date combination (daily snapshot)
Data Quality Issues	~4% NULL reorder_point values, ~6% stock_on_hand inconsistencies (negative values), ~3% holding_cost_rate outliers, ~5% missing vendor lead time, ~2% duplicate inventory_id records

Column Data Dictionary — Inventory.csv

Column Name	Description	Datatype	Business Meaning	Allowed Values	Validation Rules	Null Handling
inventory_id	Unique inventory snapshot ID	VARCHAR(15)	Primary key for inventory records	INV-XXXXXXXXXX	Unique, non-null	NOT NULL — reject
snapshot_date	Date of inventory snapshot	DATE	Time-series analysis baseline	2021-01-01 to 2024-12-31	Valid date, not future	NOT NULL
product_id	Product identifier	VARCHAR(12)	Links to Orders dataset	PRD-XXXXXX format	Must exist in orders.csv products	NOT NULL
product_name	Product description	VARCHAR(150)	Human-readable product label	Any string	Non-empty	NOT NULL
product_category	Product category	VARCHAR(50)	Category-level inventory analytics	5 categories	Must match master list	NOT NULL

sku_code	Stock Keeping Unit code	VARCHAR(20)	Unique product-warehouse variant ID	SKU-XXXX-XX format	Unique per warehouse	NOT NULL
warehouse_id	Warehouse identifier	VARCHAR(10)	Location of this inventory	WH-01 through WH-14	Must be valid warehouse	NOT NULL
warehouse_location	City/country of warehouse	VARCHAR(80)	Geographic inventory distribution	14 global locations	Must match warehouse master	NOT NULL
stock_on_hand	Current units in stock	INTEGER	Primary inventory level metric	>= 0 (negatives = issue)	Must be >= 0; negatives flagged	NOT NULL — 0 if NULL
units_reserved	Units reserved for pending orders	INTEGER	Available stock calculation	>= 0	Must be <= stock_on_hand	NOT NULL — 0 if NULL
available_stock	stock_on_hand - units_reserved	INTEGER	True available inventory	>= 0	Derived; recalculate if inconsistent	Recalculate if NULL
reorder_point	Trigger level for new purchase order	DECIMAL(10,2)	Procurement trigger metric	>= 0	Based on lead time * avg demand	4% NULL — impute from formula
reorder_quantity	Units per standard replenishment order	INTEGER	Procurement order sizing	>= 1	Economic order quantity based	NOT NULL — use historical avg
safety_stock_units	Buffer stock for demand uncertainty	INTEGER	Stockout risk buffer	>= 0	Safety stock formula applied	NOT NULL — calculate if NULL
max_stock_level	Maximum stock capacity for warehouse	INTEGER	Overstock detection threshold	>= reorder_point	Must be > reorder_point	NOT NULL
days_of_supply	Estimated days until stockout	DECIMAL(8,2)	Urgency metric for replenishment	>= 0	stock_on_hand / avg_daily_demand	NULL if demand = 0 — OK
vendor_id	Primary supplying vendor	VARCHAR(12)	Links to Vendors dataset	VND-XXXXXX format	Must exist in vendors.csv	NOT NULL
vendor_lead_time_days	Expected days from PO to receipt	INTEGER	Reorder timing calculation	1 to 120 days	Positive integer	5% NULL — use vendor avg

last_receipt_date	Date of last inventory receipt	DATE	Stock age calculation baseline	Valid date	Must be <= snapshot_date	NULL if no receipts — OK
last_issue_date	Date of last stock issue/sale	DATE	Demand recency metric	Valid date	Must be <= snapshot_date	NULL if no issues — OK
stock_age_days	Days since last movement	INTEGER	Slow-moving stock identification	>= 0	snapshot_date - last_movement_date	Recalculate if NULL
stock_classification	ABC / XYZ classification label	VARCHAR(10)	Inventory management tier	A/B/C combined with X/Y/Z	Must be valid combination	Derive from consumption data if NULL
holding_cost_rate_pct	Monthly holding cost as % of value	DECIMAL(5,2)	Working capital cost metric	0.5% to 8.0%	Flag outliers > 5%	NULL = use warehouse default rate
unit_cost_usd	Cost per unit to GlobeNex	DECIMAL(10,2)	Inventory valuation	> 0	Must be positive	NOT NULL — reject if NULL
total_inventory_value_usd	stock_on_hand * unit_cost_usd	DECIMAL(15,2)	Working capital valuation	>= 0	Derived; validate against components	Recalculate if NULL
stockout_flag	1 if stockout occurred on this date	INTEGER (0/1)	Stockout event tracking for ML	0 or 1 only	Binary flag — no other values	NOT NULL — 0 if NULL

DATASET 3: VENDORS.CSV — COMPLETE METADATA

Dataset Name	Vendors Dataset
File Name	vendors.csv
Total Rows	22,000+ records
Total Columns	30 columns
Date Range	January 2021 to December 2024 (monthly vendor performance snapshots)
Primary Key	vendor_record_id — format: VR-XXXXXXXX (unique per vendor per month)
Vendor Master Key	vendor_id — format: VND-XXXXXX (1,200+ unique vendors)
Business Purpose	Monthly vendor performance records capturing procurement metrics, quality scores, financial risk indicators, lead time performance, and contract compliance. Powers the Vendor Risk Intelligence Score (VRIS) model.
Grain	One row per vendor per month (monthly performance snapshot)
Data Quality Issues	~8% NULL quality_score, ~12% NULL financial_stability_score, ~5% lead_time inconsistencies, ~3% invalid country codes, ~7% contract_expiry_date in past without renewal flag

Column Data Dictionary — Vendors.csv

Column Name	Description	Datatype	Business Meaning	Allowed Values	Validation Rules	Null Handling
vendor_record_id	Unique record ID (PK)	VARCHAR(15)	Primary key for this monthly snapshot	VR-XXXXXXXX format	Unique per vendor-month	NOT NULL
vendor_id	Vendor master identifier	VARCHAR(12)	Cross-dataset join key	VND-XXXXXX format	Must be consistent across months	NOT NULL
vendor_name	Legal vendor company name	VARCHAR(150)	Vendor identification in reports	Any string	Non-empty	NOT NULL
vendor_country	Vendor headquarters country	VARCHAR(50)	Geographic risk assessment	Valid country names	ISO 3166 standard	NOT NULL
vendor_region	Geographic region classification	VARCHAR(30)	Regional procurement analytics	APAC, EMEA, Americas, MEA	Must match master list	Derive from country if NULL

vendor_tier	Procurement tier classification	VARCHAR(10)	Strategic importance level	Tier-1, Tier-2, Tier-3	Must match master list	NOT NULL — default Tier-3
vendor_category	Primary supply category	VARCHAR(60)	Category management analytics	Industrial Parts, Electronics, Raw Materials, Packaging, Services, Logistics	Must match master list	NOT NULL
snapshot_month	Month of this performance record	DATE (YYYY-MM-01)	Time-series analysis	2021-01 to 2024-12	First day of month format	NOT NULL
on_time_delivery_rate	% orders delivered on time (this month)	DECIMAL(5,2)	Primary delivery performance KPI	0.00 to 100.00	Percentage — flag < 70%	NOT NULL — flag if NULL
quality_acceptance_rate	% of deliveries meeting quality spec	DECIMAL(5,2)	Quality performance metric	0.00 to 100.00	Flag if < 85%	8% NULL — impute from 3M avg
quality_rejection_count	Number of rejected deliveries this month	INTEGER	Absolute quality count metric	>= 0	Positive integer or 0	NOT NULL — 0 if NULL
defect_rate_pct	% of units with defects	DECIMAL(5,3)	Manufacturing quality indicator	0.000 to 15.000	Flag if > 3%	NOT NULL — flag if NULL
lead_time_avg_days	Average actual lead time this month	DECIMAL(8,2)	Lead time performance vs. SLA	> 0	Should match contract ±20%	5% NULL — use contract lead time
lead_time_variance_days	Std dev of lead times this month	DECIMAL(8,2)	Lead time predictability metric	>= 0	High variance = reliability risk	Recalculate if NULL
contract_lead_time_days	Lead time committed in contract	INTEGER	SLA benchmark for comparison	1 to 120 days	Must match contract on file	NOT NULL — use vendor class default
procurement_spend_usd	Total spend with vendor this month	DECIMAL(15,2)	Spend analytics and concentration	> 0	Must be positive	NOT NULL

purchase_order_count	Number of POs issued this month	INTEGER	Transaction volume metric	>= 1	Positive integer	NOT NULL
invoice_accuracy_rate	% of invoices without discrepancies	DECIMAL(5,2)	Procure-to-pay efficiency	0 to 100	Flag if < 90%	NOT NULL — flag if NULL
payment_terms_days	Net payment terms (days)	INTEGER	Working capital management	Net-15 to Net-120	Must be in approved terms list	NOT NULL — use contract terms
days_payable_outstanding	Actual DPO for this vendor this month	DECIMAL(8,2)	Payables performance metric	>= 0	Compare vs. payment_terms_days	Recalculate if NULL
contract_start_date	Contract effective date	DATE	Contract lifecycle management	Valid date before snapshot_month	Must be valid date	NOT NULL
contract_expiry_date	Contract expiration date	DATE	Renewal risk management	Valid date	Flag if within 90 days of snapshot	NULL = no fixed expiry — flag
concentration_risk_pct	% of category spend with this vendor	DECIMAL(5,2)	Single-vendor dependency risk	0.00 to 100.00	Flag if > 40% for Tier-1 category	Recalculate from spend data if NULL
financial_stability_score	Third-party credit/stability rating (0-100)	DECIMAL(5,2)	Financial viability risk indicator	0.00 to 100.00	Flag if < 40	12% NULL — use sector median
esg_compliance_score	Environmental, Social, Governance score	DECIMAL(5,2)	Regulatory and reputational risk	0.00 to 100.00	Flag if < 50	NULL = not assessed — flag
past_disruption_count	Number of supply disruptions in last 12 months	INTEGER	Historical disruption risk	>= 0	Count of disruption events	NOT NULL — 0 if NULL
active_dispute_flag	1 if active commercial dispute exists	INTEGER (0/1)	Relationship risk indicator	0 or 1	Binary flag	NOT NULL — 0 if NULL
preferred_vendor_flag	1 if designated as preferred vendor	INTEGER (0/1)	Procurement preference tier	0 or 1	Binary flag	NOT NULL — 0 if NULL
vris_score	Vendor Risk Intelligence Score (composite)	DECIMAL(5,2)	PRIMARY ML target / business KPI	0.00 to 100.00 (100 = low risk)	Weighted composite — recalculate	Compute from

						components if NULL
risk_category	VRIS-derived risk label	VARCHAR(20)	Risk tier for executive reporting	Low, Medium, High, Critical	Derived from vris_score bands	Derive from vris_score if NULL

DATASET 4: LOGISTICS.CSV — COMPLETE METADATA

Dataset Name	Logistics Dataset
File Name	logistics.csv
Total Rows	85,000+ records
Total Columns	32 columns
Date Range	January 2021 to December 2024
Primary Key	shipment_id — format: SHP-XXXXXXXX
Foreign Keys	order_id (→ Orders), carrier_id, route_id, warehouse_id
Business Purpose	Tracks individual shipment records across GlobeNex's global logistics network. Captures carrier performance, route details, delivery timelines, delay causes, weather impact, and cost metrics. Powers the Delivery Delay Prediction and Route Risk models.
Grain	One row per shipment (a single order_id may have multiple shipments for split deliveries)
Data Quality Issues	~4% NULL actual_delivery_date for delivered shipments, ~6% fuel_cost_usd outliers (10x normal), ~3% duplicate shipment_id from OTM integration errors, ~7% missing delay_reason for late shipments, ~5% weather_impact flag inconsistencies

Column Data Dictionary — Logistics.csv (Selected Key Columns)

Column Name	Description	Datatype	Business Meaning	Allowed Values	Validation Rules	Null Handling
shipment_id	Unique shipment identifier	VARCHAR(15)	Primary key	SHP-XXXXXXXX	Unique, non-null	NOT NULL
order_id	Linked order identifier	VARCHAR(15)	FK to Orders dataset	ORD-XXXXXXXX	Must exist in orders.csv	NOT NULL
shipment_date	Date shipment was dispatched	DATE	Lead time calculation start	Valid date	Must be <= actual_delivery_date	NOT NULL
promised_delivery_date	Committed delivery date	DATE	OTD calculation baseline	Valid date	>= shipment_date	NOT NULL
actual_delivery_date	Actual delivery date	DATE	OTD outcome metric	Valid date or NULL	NULL for in-transit	NULL if in-transit — OK; NULL if delivered — flag

delivery_status	Current shipment status	VARCHAR(30)	Fulfillment tracking	In-Transit, Delivered, Delayed, Failed, Returned	Normalize case	NOT NULL
carrier_id	Logistics carrier identifier	VARCHAR(12)	Carrier analytics	CAR-XXXXXX format	Must exist in carrier master	NOT NULL
carrier_name	Carrier company name	VARCHAR(100)	Carrier performance reporting	Any string	Non-empty	NOT NULL
carrier_type	Mode of transportation	VARCHAR(30)	Modal analytics	Air, Ocean, Road, Rail, Multimodal	Must match master list	NOT NULL
origin_country	Country of shipment origin	VARCHAR(50)	Route analytics	Valid country	ISO 3166	NOT NULL
destination_country	Country of delivery destination	VARCHAR(50)	Route analytics	Valid country	ISO 3166	NOT NULL
route_id	Predefined route identifier	VARCHAR(12)	Route performance analytics	RTE-XXXXXX format	Must exist in route master	NOT NULL
transit_days_planned	Planned transit duration	INTEGER	Benchmark for delay calculation	> 0	Positive integer	NOT NULL
transit_days_actual	Actual transit duration	INTEGER	Delay measurement	> 0 or NULL	NULL if in-transit	NULL if in-transit — OK
delay_days	Days late vs. promised date	INTEGER	SLA breach indicator; ML target	Any integer (negative = early)	transit_actual - transit_planned	Recalculate if NULL
delay_flag	1 if delivered late, 0 if on-time	INTEGER (0/1)	Binary ML target for delay model	0 or 1	Binary — derive from delay_days > 0	NOT NULL — derive
delay_reason	Primary cause of delay	VARCHAR(100)	Root cause analytics	Weather, Customs, Carrier Issue, Traffic, Port Congestion, Mechanical Failure, Documentation, Unknown	Only if delay_flag = 1	7% NULL for delayed shipments — flag
weather_impact_flag	1 if weather was a contributing delay factor	INTEGER (0/1)	Weather risk analysis	0 or 1	Binary flag	NOT NULL — 0 if NULL

weather_severity	Severity of weather impact	VARCHAR(20)	Weather risk scoring	None, Low, Moderate, Severe, Extreme	Only if weather_impact_flag = 1	NULL if no weather — OK
customs_clearance_days	Days in customs processing	INTEGER	Cross-border delay analysis	>= 0	0 for domestic shipments	NOT NULL — 0 for domestic
shipment_weight_kg	Total shipment weight	DECIMAL(10,2)	Cost calculation base	> 0	Positive value	NOT NULL
shipment_volume_cbm	Total volume in cubic meters	DECIMAL(10,3)	Volumetric weight pricing	> 0	Positive value	NOT NULL
freight_cost_usd	Base freight charge	DECIMAL(12,2)	Primary logistics cost metric	> 0	Positive value	NOT NULL
fuel_surcharge_usd	Fuel surcharge applied	DECIMAL(10,2)	Fuel cost analytics	>= 0	Must be >= 0; flag > 3x freight	NULL = 0 — set to 0
customs_duty_usd	Customs duty and import taxes	DECIMAL(12,2)	Cross-border cost analytics	>= 0	0 for domestic shipments	NOT NULL — 0 for domestic
total_logistics_cost_usd	Sum of all logistics cost components	DECIMAL(14,2)	Total cost for profitability model	> 0	Must equal sum of components	Recalculate if NULL
cost_per_kg_usd	Logistics cost efficiency metric	DECIMAL(8,4)	Cost efficiency benchmarking	> 0	Derived from total_cost / weight	Recalculate if NULL
carrier_rating_score	GlobeNex internal carrier performance score	DECIMAL(5,2)	Carrier selection optimization	0.00 to 100.00	Updated monthly	NULL = new carrier — use initial score 50
last_port_scan	Last GPS/scan checkpoint location	VARCHAR(100)	Real-time tracking (simulated)	City, Country format	Non-empty for in-transit	NULL for delivered — OK
co2_emissions_kg	Estimated CO2 emissions for this shipment	DECIMAL(10,2)	ESG/sustainability reporting	> 0	Based on carrier type and distance	NOT NULL — estimate if NULL
insurance_cost_usd	Cargo insurance premium	DECIMAL(10,2)	Risk cost analytics	>= 0	% of shipment value	NULL = 0 — set to 0

shipment_notes	Free-text notes from logistics ops	VARCHAR(500)	Qualitative delay context	Any string	No validation — text field	NULL = no notes — OK

DATASET 5: FINANCIALS.CSV — COMPLETE METADATA

Dataset Name	Financial Records Dataset
File Name	financials.csv
Total Rows	65,000+ records
Total Columns	28 columns
Date Range	January 2021 to December 2024 (weekly financial snapshots)
Primary Key	finance_record_id — format: FIN-XXXXXXXXXX
Foreign Keys	order_id (→ Orders), vendor_id (→ Vendors), shipment_id (→ Logistics)
Business Purpose	Captures weekly financial records integrating revenue, procurement costs, operational expenses, working capital metrics, cash flow positions, and financial risk indicators. Powers cash flow forecasting, working capital optimization, and profitability analytics.
Grain	One row per transaction/event (order payment, vendor invoice, operational expense)
Data Quality Issues	~6% NULL working_capital_ratio, ~4% cash_flow_usd sign inconsistencies, ~5% duplicate records from AS/400 reconciliation, ~8% missing financial_risk_score, ~3% procurement_cost exceeding order_value

Column Data Dictionary — Financials.csv (Key Columns)

Column Name	Description	Datatype	Business Meaning	Allowed Values	Validation Rules	Null Handling
finance_record_id	Unique financial record ID	VARCHAR(15)	Primary key	FIN-XXXXXXXXXX	Unique, non-null	NOT NULL
record_date	Date of financial record	DATE	Financial timeline reference	2021-01-01 to 2024-12-31	Valid date	NOT NULL
fiscal_week	Fiscal week number (1-52)	INTEGER	Weekly financial aggregation	1 to 53	Integer in valid range	Derive from record_date if NULL
fiscal_quarter	Fiscal quarter (Q1-Q4)	VARCHAR(5)	Quarterly analytics	Q1, Q2, Q3, Q4	Must match master list	Derive from record_date if NULL
fiscal_year	Fiscal year	INTEGER	Annual trend analysis	2021 to 2024	Valid year	Derive from record_date if NULL

order_id	Linked order (if applicable)	VARCHAR(15)	FK to Orders dataset	ORD-XXXXXXXX or NULL	Exists in orders.csv or NULL	NULL for non-order expenses — OK
vendor_id	Linked vendor (if applicable)	VARCHAR(12)	FK to Vendors dataset	VND-XXXXXX or NULL	Exists in vendors.csv or NULL	NULL for non-vendor transactions — OK
shipment_id	Linked shipment (if applicable)	VARCHAR(15)	FK to Logistics dataset	SHP-XXXXXXXX or NULL	Exists in logistics.csv or NULL	NULL for non-logistics records — OK
transaction_type	Type of financial transaction	VARCHAR(40)	Classification for P&L mapping	Revenue, COGS, Procurement, Logistics Cost, Operational Expense, SLA Penalty, Interest Expense	Must match master list	NOT NULL
revenue_usd	Revenue recognized this record	DECIMAL(15,2)	Top-line revenue tracking	>= 0	0 for expense records	NOT NULL — 0 for non-revenue
procurement_cost_usd	Cost of procurement this record	DECIMAL(15,2)	COGS and procurement analytics	>= 0	0 for revenue records	NOT NULL — 0 for non-procurement
logistics_cost_usd	Freight and logistics costs	DECIMAL(12,2)	Logistics cost analytics	>= 0	Should match logistics.csv total	NOT NULL — 0 if no logistics
operational_cost_usd	Overhead, warehousing, labor costs	DECIMAL(12,2)	Operational efficiency analytics	>= 0	Positive value	NOT NULL
sla_penalty_usd	Penalty paid for SLA breaches	DECIMAL(10,2)	SLA cost impact analytics	>= 0	0 if no breach	NOT NULL — 0 if NULL
gross_profit_usd	Revenue minus COGS	DECIMAL(15,2)	Profitability measurement	Any value (can be negative)	revenue - COGS	Recalculate if NULL

ebitda_usd	Earnings before interest, tax, depreciation	DECIMAL(15,2)	Operating performance metric	Any value	Derived metric	Calculate from components if NULL
accounts_receivable_usd	Outstanding receivables this period	DECIMAL(15,2)	Working capital — cash inflow risk	>= 0	Positive value	NOT NULL — 0 if NULL
accounts_payable_usd	Outstanding payables this period	DECIMAL(15,2)	Working capital — cash outflow obligation	>= 0	Positive value	NOT NULL — 0 if NULL
inventory_value_usd	Value of on-hand inventory	DECIMAL(15,2)	Working capital — inventory component	>= 0	Should reconcile with inventory.csv	NOT NULL
working_capital_usd	AR + Inventory - AP (net)	DECIMAL(15,2)	Working capital position	Any value (negative = issue)	working_capital = AR + Inv - AP	6% NULL — recalculate
working_capital_ratio	Current ratio (current assets / current liabilities)	DECIMAL(8,4)	Liquidity risk indicator	> 0	Flag if < 1.0 (liquidity warning)	6% NULL — calculate if possible
cash_flow_usd	Net cash flow for this period	DECIMAL(15,2)	Treasury management metric	Any value	Positive = inflow, Negative = outflow	NOT NULL — 0 if NULL
cash_position_usd	Cumulative cash balance	DECIMAL(15,2)	Treasury cash management	Any value	Running sum of cash_flow_usd	Recalculate if NULL
days_sales_outstanding	Average DSO for this period	DECIMAL(8,2)	Collections efficiency metric	>= 0	Industry benchmark: 45-60 days	NOT NULL — flag if NULL
days_payable_outstanding	Average DPO for this period	DECIMAL(8,2)	Payables management metric	>= 0	Target: 60-90 days for GlobeNex	NOT NULL — flag if NULL
cash_conversion_cycle	DSO + DIO - DPO	DECIMAL(8,2)	Working capital efficiency (days)	Any value (lower = better)	Derived metric; target < 45 days	Recalculate from components
financial_risk_score	Composite financial risk score (0-100)	DECIMAL(5,2)	Financial health indicator; ML target	0.00 to 100.00 (100 = high risk)	Weighted composite	8% NULL — compute from components
risk_flag	Binary financial risk alert	INTEGER (0/1)	Alert trigger for treasury team	0 = normal, 1 = risk alert	1 if risk_score > 70 or WC_ratio < 1.0	Derive if NULL

--- END OF DOCUMENT 2: DATASET DESIGN & METADATA DOCUMENT ---

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